



SAL ENGINEERING AND TECHNICAL INSTITUTE

CE, IT, CSE & ICT Department

**WEB APPLICATION DEVELOPMENT
(BE05000281)**

Question Bank

Year: 2026-2027

Prepared By:

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CHAPTER 1: Introduction to Web Technologies (CO-1)			
Sr No.	Question	CO Mapping	Marks
1	Define Web Browser. What is its role in web communication? (WD W25)	CO-1	3
2	Define WWW. Explain its evolution briefly. (WD W21,W22)	CO-1	3
3	Explain and give one example for: (1) Website (2) URL (3) Domain. (WD W24)	CO-1	3
4	Define Web Technology. State its applications.	CO-1	3
5	Differentiate Static Website and Dynamic Website.	CO-1	3
6	Define Client–Server Architecture with a suitable example.	CO-1	3
7	What is Frontend? List any four Frontend technologies.	CO-1	3
8	What is Backend? State its functions in a web application.	CO-1	3
9	What is a Database? Explain its role in a web application.	CO-1	3
10	List the advantages of Web Technologies.	CO-1	3
11	Explain the role of Web Technologies in Banking and Finance.	CO-1	3
12	Draw and explain the architecture of the Web Browser. (WD S25,W22)	CO-1	4
13	Differentiate between Client-Side Scripting and Server-Side Scripting. (WD W22,S23)	CO-1	4
14	Explain Client–Server Architecture with a neat diagram.	CO-1	4
15	Differentiate Frontend, Backend and Database.	CO-1	4
16	Compare Static Websites and Dynamic Websites.	CO-1	4
17	Explain the evolution of websites from Web 1.0 to modern web applications.	CO-1	4
18	Describe the working of a web application using Client, Server and Database.	CO-1	4
19	Explain the role of Web Technologies in Banking and Finance with suitable examples.	CO-1	4
20	Explain Client–Server Architecture in detail with a neat diagram. Describe the role of Client, Server and Database.	CO-1	7
21	Explain the evolution of websites. Differentiate Web 1.0, Web 2.0 and Modern Web Applications.	CO-1	7
22	Differentiate Static and Dynamic Websites. Explain their advantages, disadvantages and applications.	CO-1	7
23	Explain Frontend, Backend and Database Architecture with suitable examples.	CO-1	7
24	Discuss the role of Web Technologies in Banking and Finance. Explain how modern banking systems use web technologies.	CO-1	7
25	Explain the complete working of a dynamic website using Client, Server and Database with a neat diagram.	CO-1	7

CHAPTER 2: HTML & CSS Fundamentals (CO-2)			
Sr No.	Question	CO Mapping	Marks
1	Define Bootstrap for CSS. Why do we use it? (WD W25,S25)	CO-2	3
2	What is Cascading Style Sheet (CSS)? Write its full form. (WD W21,W22)	CO-2	3
3	Describe Selectors in CSS. (AWP W25)	CO-2	3
4	Write HTML code to take birth date input between 01-01-2005 to 31-12-2024. (WD S26,S22)	CO-2	3

5	Use CSS animations so that a button changes color and scales up when hovered. Write code. (WD S26)	CO-2	3
6	What is responsive design? Explain with suitable example and media queries. (AWP W24,S26)	CO-2	4
7	Compare Inline, Embedded and External style sheets with example. (WD W25,W22)	CO-2	4
8	Explain CSS pseudo-class selector with example. (WD W25,W22)	CO-2	4
9	Explain ordered and unordered lists in HTML with example. (WD W25,W22)	CO-2	4
10	Explain different types of CSS with suitable example. (AWP S26,W25)	CO-2	4
11	What is Flexbox Layout? Explain with CSS example. (WD Syllabus)	CO-2	4
12	Explain the CSS Box Model. Define margin, padding and border. (WD S23)	CO-2	4
13	Explain class and ID selector in CSS with example. (WD W25)	CO-2	4
14	Explain CSS Wildcard Selectors (*, ^ and \$) with example. (WD W22)	CO-2	4
15	Write HTML & CSS code to create Horizontal and Vertical Menu. (WD S22)	CO-2	4
16	What is CSS? Explain different types of CSS. List out advantages and disadvantages of CSS. (WD S24,W21)	CO-2	7
17	Create an HTML form for student registration using textbox, checkbox, radio button, combo box, file upload, submit and reset button. (WD W24,W21)	CO-2	7
18	Write an HTML code to create a student registration form for college admission with validations: all fields compulsory, mobile 10 digits, DOB range, email and pin code validation. (WD S26)	CO-2	7
19	What is HTML table? Explain table element with necessary attributes and write code to produce a given table layout. (WD W25)	CO-2	7
20	Explain how basic and nested tables are created using HTML. (WD W23)	CO-2	7
21	Write HTML and CSS code: (i) Set background color for active link to 'blue'. (ii) Set unordered list style to 'circle'. (iii) Set 'animation.gif' as background image. (WD W25)	CO-2	7
22	Write CSS code to implement: (1) Ordered list with red border, padding, font-size 20. (2) Use flower.jpg as background. (3) Change background on text box focus. (4) Create fixed position link at bottom. (WD S26,S22)	CO-2	7
23	Design a responsive personal portfolio webpage using HTML and CSS. Include navbar, about, skills and contact sections. (WD Lab)	CO-2	7
24	Write HTML code to create your class time table using table tag with proper formatting using CSS. (WD S25,W23)	CO-2	7

CHAPTER 3: JavaScript Fundamentals (CO-3)			
Sr No.	Question	CO Mapping	Marks
1	Write a JavaScript function to calculate the sum of three given numbers. (WD S25)	CO-3	3
2	Discuss JavaScript alert, prompt, and confirm with example. (WD S25,W21)	CO-3	3
3	Write a JavaScript to demonstrate alert, confirm, and prompt box. (WD W24)	CO-3	3
4	List and explain built-in objects of JavaScript with one example each. (WD W24)	CO-3	3
5	Discuss array in JavaScript with example. (WD S25)	CO-3	3
6	Define term: DOM. State the usefulness of DOM in JavaScript. (AWP S25,W25)	CO-3	3

7	Describe the role of callbacks in JavaScript functions with a simple code snippet. (WD S26)	CO-3	3
8	Explain DOM in detail with different ways to access HTML element value. (AWP W25,S26)	CO-3	4
9	Write a JavaScript program to demonstrate use of Array functions: indexOf(), lastIndexOf(), includes(), sort(), reverse(). (AWP W25)	CO-3	4
10	Explain event handling in JavaScript with proper example. (WD S23,S26)	CO-3	4
11	Write a JavaScript code to change image size on mouseover and revert on mouseout. (WD S26,S22)	CO-3	4
12	Explain callback function in JavaScript with suitable example. (WD W25,S24)	CO-3	4
14	Explain asynchronous programming in JavaScript. Simulate using setTimeout and a callback. (AWP W24,S26)	CO-3	4
15	Explain Events in JavaScript with suitable example. (WD W25)	CO-3	4
16	Using JavaScript, create a simple calculator for addition, subtraction, multiplication and division. (WD W24)	CO-3	4
17	Write a javascript function to find maximum value among three values entered by user. (WD S25,W21)	CO-3	4
18	Explain pop-up boxes in JavaScript (Alert, Prompt, Confirm) with example. (WD W22,W23)	CO-3	7
19	Write JavaScript code to display the text 'Web Programming' with increasing font size in intervals of 20ms in blue. Stop when font size reaches 40px. (WD W25)	CO-3	7
20	Prepare a JavaScript solution to validate a registration form including email, password (6-12 chars), username (not starting with _, @ or number), and all fields not blank. (WD W25,W22,W21)	CO-3	7
21	Write JavaScript to find whether the entered number by user is Prime or Not. (WD W22,W21)	CO-3	7
22	Write JavaScript function to display prime numbers among 1 to 100 using loop. (WD S25)	CO-3	7
23	Write JavaScript code to parse a JSON file containing student data and display it in an HTML table. (WD S26)	CO-3	7
24	Write a JavaScript program to validate user account data from JSON and display Login Successful or Invalid Username/Password. (WD S26)	CO-3	7
25	Create a registration form using JavaScript on one button click using: createElement(), appendChild(), setAttribute(). (WD W24)	CO-3	7
26	Write JavaScript code to read user account data stored in JSON. Design HTML form for Username and Password. Validate credentials against stored JSON data. (WD S26)	CO-3	7
27	Write JavaScript code: create account object with sub-objects (Saving, Loan, Current Account) with properties. Display all using for-in loop. (WD S26)	CO-3	7
28	Write JavaScript code to enter 10 numbers via prompt (comma separated), find Maximum, Minimum and Average and display via alert. (WD W25)	CO-3	7

CHAPTER 4: APIs & HTTP Communication (CO-3)

Sr No.	Question	CO Mapping	Marks
1	Define API. State any four advantages of using APIs.	CO-3	3
2	What is Client–Server Communication? Explain with a simple example.	CO-3	3

3	Define HTTP Request and HTTP Response.	CO-3	3
4	What are HTTP Status Codes? Give any three commonly used status codes with meaning.	CO-3	3
5	Define Query Parameters with suitable example.	CO-3	3
6	What is Request Body? State its purpose.	CO-3	3
7	Define CRUD Operations. Expand the term CRUD.	CO-3	3
8	What is JSON? State its advantages over XML.	CO-3	3
9	Define XML. List its features.	CO-3	3
10	What is Fetch API? State its advantages.	CO-3	3
11	Differentiate between Public API and Private API.	CO-3	3
12	List the commonly used HTTP Methods.	CO-3	3
13	Explain the Client–Server Communication process with a neat diagram.	CO-3	4
14	Explain the structure of an HTTP Request with suitable example.	CO-3	4
15	Explain the structure of an HTTP Response with suitable example.	CO-3	4
16	Explain different HTTP Methods (GET, POST, PUT, DELETE) with examples.	CO-3	4
17	Explain HTTP Status Code categories (1xx, 2xx, 3xx, 4xx, 5xx).	CO-3	4
18	Differentiate JSON and XML.	CO-3	4
19	Explain Query Parameters and Request Body with suitable examples.	CO-3	4
20	Explain CRUD Operations using a Student Management System example.	CO-3	4
21	Explain the Fetch API syntax with suitable example.	CO-3	4
22	Write JavaScript code to fetch data from a public API using Fetch API.	CO-3	4
23	Explain the advantages and limitations of Fetch API.	CO-3	4
24	Differentiate HTTP GET and POST methods.	CO-3	4
25	What is HTTP? Explain HTTP Protocol Header, HTTP Request and HTTP Response with example. (WD W25, W22, W21)	CO-3	7
26	Explain HTTP Request and HTTP Response in detail with suitable examples.	CO-3	7
27	Explain all HTTP Methods (GET, POST, PUT, DELETE) with practical examples.	CO-3	7
28	Explain HTTP Status Codes in detail with suitable examples for each category.	CO-3	7
29	Explain CRUD Operations. Write JavaScript code using Fetch API to perform GET and POST operations.	CO-3	7
30	Write JavaScript code to retrieve user information from a Public API using Fetch API and display it in an HTML table.	CO-3	7
31	Explain JSON and XML in detail. Compare both formats with suitable examples.	CO-3	7
32	Write JavaScript code to consume a Weather/Public API using Fetch API and display the received data on a webpage.	CO-3	7
33	Develop a webpage that fetches student data from a JSON file/API and displays it in a tabular format using Fetch API.	CO-3	7
34	Write JavaScript code to perform CRUD operations on a REST API using Fetch API.	CO-3	7
35	Explain the complete working of Client–Server Communication while accessing a Public API with the help of HTTP Request and Response.	CO-3	7
	Practical Oriented Questions	CO-3	
36	Write JavaScript code to fetch and display a list of users from a public REST API.	CO-3	7
37	Write JavaScript code to send registration data to an API using the POST method.	CO-3	7

38	Write JavaScript code to update an existing record using the PUT method.	CO-3	7
39	Write JavaScript code to delete a record using the DELETE method.	CO-3	7
40	Create an HTML page that accepts a user ID and fetches the corresponding user details from a public API.	CO-3	7
41	Write JavaScript code to handle API errors (404, 500) while fetching data using Fetch API.	CO-3	7
42	Develop a webpage that searches products using query parameters and displays the results.	CO-3	7
43	Write JavaScript code to fetch JSON data asynchronously and display it in a responsive HTML table	CO-3	7

CHAPTER 5: Backend Development with Node.js (CO-4)

Sr No.	Question	CO Mapping	Marks
1	What is Node.js? List and explain its features. <i>(AWP W25,S25)</i>	CO-4	3
2	Write a Node.js code to create an HTTP server. <i>(AWP W25,S26)</i>	CO-4	3
3	Explain REST architecture in Node.js. <i>(AWP W25)</i>	CO-4	3
4	Explain error handling in Node.js with suitable example. <i>(AWP W24)</i>	CO-4	3
5	Explain core modules of Node.js. <i>(AWP S26)</i>	CO-4	3
6	What is NPM? Explain its role in Node.js. How to install any module — explain with syntax. <i>(AWP W25,S25)</i>	CO-4	3
7	Write a short note on Errors in Node.js. <i>(AWP W25)</i>	CO-4	3
8	What is Runtime Environment? Explain the role of Node.js Runtime.	CO-4	3
9	Differentiate Node.js and Traditional Server-side Programming.	CO-4	3
10	What is Express.js? State its advantages.	CO-4	3
11	Define Middleware in Express.js.	CO-4	3
12	What is Routing? Explain its purpose in Express.js.	CO-4	3
13	What is REST API? State its characteristics.	CO-4	3
14	List HTTP methods commonly used in REST APIs.	CO-4	3
15	Explain Node.js REST API in detail. <i>(AWP W24)</i>	CO-4	4
16	Explain the difference between blocking and non-blocking code using Node.js example. <i>(AWP W25)</i>	CO-4	4
17	Define Node Package Manager (NPM). Explain the process of installing, upgrading and uninstalling a module using npm. <i>(AWP W24,S26)</i>	CO-4	4
18	Explain the event-driven programming concept in Node.js with example. <i>(AWP W25)</i>	CO-4	4
19	Explain the architecture of Node.js with a neat diagram.	CO-4	4
20	Explain Event Loop in Node.js.	CO-4	4
21	Explain Express.js Routing with suitable example.	CO-4	4
22	Explain Middleware types in Express.js.	CO-4	4
23	Explain REST API Architecture.	CO-4	4
24	Differentiate GET, POST, PUT and DELETE requests.	CO-4	4
25	Explain Database Connectivity using Node.js and Express.js.	CO-4	4
26	Explain the steps to connect Node.js with MongoDB/MySQL.	CO-4	4
27	Explain Authentication and Authorization with suitable example.	CO-4	4

28	What is Express.js? Explain Routing and Middleware basics. Write a REST API using Express.js.	CO-4	7
29	Write Node.js code to print values of query string variables (Name, Mobile) from a URL. (AWP S25,W24)	CO-4	7
30	Write a Node.js code to create a server that logs visited pages in a log.txt file. (AWP W24)	CO-4	7
31	Explain API Testing using Postman. Write steps to test a REST API created with Node.js and Express.js.	CO-4	7
32	Explain Authentication and Authorization in Node.js with JWT. Write code for login endpoint.	CO-4	7
33	Write a Node.js program to create an HTTP Server and return HTML content.	CO-4	7
34	Develop a REST API using Express.js to perform CRUD operations on Student records.	CO-4	7
35	Write Express.js code demonstrating Routing and Middleware.	CO-4	7
36	Explain Database Connectivity in Node.js and write code to insert and retrieve records.	CO-4	7
37	Develop a REST API for Employee Management using Express.js and Database.	CO-4	7
38	Explain API Testing using Postman with suitable screenshots/steps.	CO-4	7
39	Develop a Student Registration API using Express.js and perform CRUD operations.	CO-4	7
40	Create a REST API to perform CRUD operations on Product data using Express.js and Database.	CO-4	7
41	Write Node.js code to connect with MongoDB/MySQL and fetch all records.	CO-4	7

CHAPTER 6: Frontend Development with React.js (CO-5)

Sr No.	Question	CO Mapping	Marks
1	What is React.js? State its advantages.	CO-5	3
2	Define JSX. State its benefits.	CO-5	3
3	What is a Component in React.js?	CO-5	3
4	Differentiate Functional Component and Class Component.	CO-5	3
5	What are Props in React.js?	CO-5	3
6	Define State in React.js.	CO-5	3
7	What is the useState Hook?	CO-5	3
8	What is the useEffect Hook?	CO-5	3
9	What is Conditional Rendering?	CO-5	3
10	What is List Rendering?	CO-5	3
11	What is Axios? Why is it used?	CO-5	3
12	What is CORS? Why is it required?	CO-5	3
13	Explain the architecture of React.js with neat diagram.	CO-5	4
14	Explain JSX with suitable example.	CO-5	4
15	Explain Functional Components with example.	CO-5	4
16	Explain Props and State with suitable examples.	CO-5	4
17	Write a React component using the useState Hook.	CO-5	4
18	Explain useEffect Hook with suitable example.	CO-5	4
19	Explain Conditional Rendering in React.js with example.	CO-5	4
20	Explain List Rendering using the map() function with example.	CO-5	4

21	Differentiate Axios and Fetch API.	CO-5	4
22	Explain CORS with suitable example.	CO-5	4
23	Write a React program to display a student list using list rendering.	CO-5	4
24	Write a React component to display user details using Props.	CO-5	4
25	Explain React.js in detail. Discuss its features and advantages.	CO-5	7
26	Develop a React application demonstrating Functional Components, Props and State.	CO-5	7
27	Write a React program using the useState Hook to implement a Counter application.	CO-5	7
28	Write a React program using useEffect to fetch data from a public API and display it.	CO-5	7
29	Develop a React application that displays a list of students using List Rendering.	CO-5	7
30	Write a React application demonstrating Conditional Rendering for Login and Logout functionality.	CO-5	7
31	Develop a React application that consumes a public REST API using Axios and displays the data in a table.	CO-5	7
32	Write a React application that fetches product details using the Fetch API and displays them using cards.	CO-5	7
33	Develop a React application to perform CRUD operations by consuming a REST API using Axios.	CO-5	7
34	Explain CORS in detail. Describe how CORS issues occur while integrating React applications with backend APIs and how they can be resolved.	CO-5	7

CHAPTER 7: Deployment & Modern Web Concepts (CO-5)

Sr No.	Question	CO Mapping	Marks
1	What is Cloud Computing? Explain its types (IaaS, PaaS, SaaS).	CO-5	3
2	What is Hosting? Explain domain management basics.	CO-5	3
3	What is CAPTCHA? Explain its concept with suitable example.	CO-5	3
4	What is Web Hosting? State its advantages.	CO-5	3
5	Differentiate Free Hosting and Paid Hosting.	CO-5	3
6	What is a Domain Name? Explain its structure.	CO-5	3
7	What is DNS? State its role in Domain Management.	CO-5	3
8	What is SEO? State its objectives.	CO-5	3
9	List different types of CAPTCHA.	CO-5	3
10	What is Cloud Computing? State its characteristics.	CO-5	3
11	Differentiate IaaS, PaaS and SaaS.	CO-5	3
12	What is SSL Certificate? Why is it required in web hosting?	CO-5	3
13	What is SEO? Explain SEO introduction and best practices for web developers.	CO-5	4
14	Explain different types of Web Hosting with suitable examples.	CO-5	4
15	Explain Domain Registration and Domain Management process.	CO-5	4
16	Explain the working of DNS with a neat diagram.	CO-5	4
17	Explain the importance of SEO for modern websites.	CO-5	4
18	Discuss On-page SEO and Off-page SEO.	CO-5	4
19	Explain different types of CAPTCHA with suitable examples.	CO-5	4
20	Explain Cloud Computing deployment models (Public, Private, Hybrid).	CO-5	4

21	Differentiate Traditional Hosting and Cloud Hosting.	CO-5	4
22	Explain the advantages and limitations of Cloud Computing.	CO-5	4
23	Explain introduction to Cloud Concepts. What are the advantages of cloud deployment for web applications?	CO-5	7
24	Explain Hosting Basics and Deployment process. How do you deploy a React.js application on a cloud platform?	CO-5	7
25	Explain Web Hosting in detail. Discuss different types of hosting with their advantages and disadvantages.	CO-5	7
26	Explain Domain Management in detail. Describe the complete process of registering and managing a domain.	CO-5	7
27	Explain SEO in detail. Discuss On-page SEO, Off-page SEO and Technical SEO with suitable examples.	CO-5	7
28	Explain CAPTCHA in detail. Discuss different CAPTCHA techniques and their applications in web security.	CO-5	7
29	Explain Cloud Computing in detail. Discuss its characteristics, deployment models and service models (IaaS, PaaS, SaaS).	CO-5	7
30	Differentiate Traditional Web Hosting and Cloud Hosting. Explain the advantages of cloud deployment for modern web applications.	CO-5	7
31	Explain the complete process of deploying a website from development to production, including hosting, domain configuration and SSL certificate setup.	CO-5	7
32	Explain the role of DNS, Domain Name, Web Hosting and Cloud Computing in deploying a modern web application.	CO-5	7

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